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Dating the Late Precambrian volcanicity of England and Wales

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Abstract: Ages for volcanism in the British Late Precambrian have been inferred from interpretations of SHRIMP zircon ages as follows: 559.3 ± 2.0 Ma for a tuff from the Beacon Hill Formation of the Charnian Supergroup; 566 ± 3 Ma for a tuff at Bardon Hill also in the Charnwood Forest area with abundant inherited grains at 590.5 ± 1.6; 566.6 ± 2.9 Ma and 555.9 ± 3.5 Ma for bentonite and tuff in the Stretton Group of the Longmyndian Supergroup; 604.7 ± 1.6 Ma for an ignimbrite of the Padarn Tuff Formation of the Arfon Group, and 572.5 ± 1.2 Ma for a tuff within the Fachwen Formation of the Arfon Group. These ages confirm that there were two major phases of volcanic activity in the English Midlands and Wales, one about 620–590 Ma and another about 575–550 Ma. This was followed in most of England and Wales by a major phase of tectonism before the Cambrian (Tommotian) transgression across the Midland Platform. Within the depocentre of the Welsh early Palaeozoic basin, however, sedimentation started within the second period of volcanism and may have continued without interruption well into the Cambrian Period. It is probable that the late phases of the Avalonian Orogeny lasted until after the beginning of the Cambrian. These new data show that the terranes of the Avalonian of England and Wales are comparable with those of eastern Canada and New England. The age estimate for the *Charnia* horizon is broadly similar to those obtained elsewhere in the world on rocks containing the earliest Ediacaran fauna and adds to the building of an extended time-scale for the global Ediacaran.

Keywords: Avalonian, Charnian, Longmyndian, Arfon Group, Ediacaran.

Since the publication of an age of 560 ± 1 Ma for the Ercall Granophyre (Tucker & Pharaoh 1991) and the dating of early Cambrian strata in other parts of the world at about 530 Ma (Compston *et al.* 1992; Isachsen *et al.* 1994; Sambridge & Compston 1994), it has been apparent that the Late Precambrian rocks of Wales and the English Midlands (Fig. 1) were subjected to a complex sequence of geological events in quite a short period of time (Wright *et al.* 1993). In order better to constrain the timing of these events, and, in particular, to determine the age of deposition of the *Charnia* fauna from the type area and to obtain data from other undated sequences, we have sought samples of tuffs having sufficient zircons for dating purposes from a number of horizons within the English and Welsh Late Precambrian. Samples which have given satisfactory data have been found in the Charnian Supergroup, the Longmyndian Supergroup and the Arfon Group. The up-to-date stratigraphy and earlier radiometric dates on these and other Late Precambrian rocks of this area are given in Pharaoh & Gibbons (1994).

Small hand specimens were collected by A.E.W. and P.T. from *in situ* exposures. They were split and hand specimens, some with thin sections, are housed in the Lapworth Museum of the School of Earth Sciences at the University of Birmingham. The remainder was processed to extract zircons using standard density methods at the Research School of Earth Sciences, ANU. Table 1 lists the field, BIRUG (Birmingham University Geology Museum) and ANU numbers for the analysed samples.

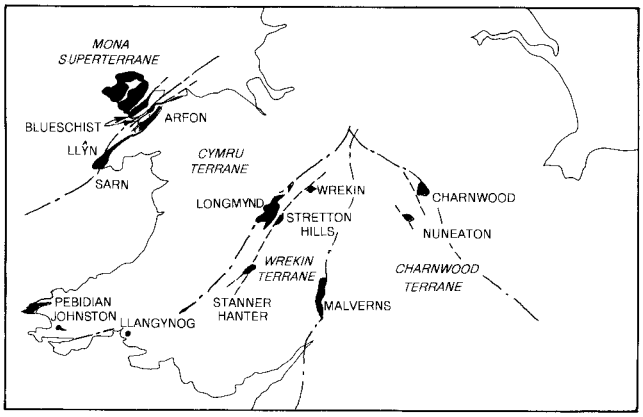


Fig. 1. Late Precambrian outcrops of Wales and the English Midlands. Dot–dash lines are terrane boundaries of Pharaoh & Carney (2000).

Correlation of Late Precambrian sequences and sample selection

The problem posed by 12 isolated inliers with differing sequences of sedimentary, volcanic and plutonic rocks and several periods of tectonism, is that petrographic, geochemical and structural similarity can never be as reliable as palaeontological or precise radiometric data for correlation purposes. Within England and Wales only two sequences have a useful fossil content and accurate age data are available for only seven events (in four inliers), and they do not date the fossiliferous horizons. Modern terrane analysis has made little advance on the correlation problems of the first attempt at plate tectonic interpretation of the area (Wright 1969) when only very imprecise age data were available. Three terranes are now recognized covering the area sampled in this study (Fig. 1) the Charnwood, Wrekin and Cymru Terranes (Gibbons & Horák 1996; Pharaoh & Carney 2000).

Table 1. *Locality details and interpreted ages of all samples in this study*

Sample no.	BIRUG no.	ANU no.	Rock type	Stratigraphic horizon	Locality	Grid ref.	Mixture modelling ages	Inherited outliers	Interpreted age of deposition
CH2	10890	Z2479 95-011	Coarse tuff matrix to slump breccia	Park Breccia Member	A50 near Markfield Charnwood	44861 31095	573.2 ± 1.0 559.3 ± 2.0 548.7 ± 1.9	573, 585, 630, 680, 700, 730, 1015, 1950	559.3 ± 2.0
CH8	10892	Z2108 95-002	Coarse tuff	Bardon Hill Complex	Bardon Hill Quarry, Charnwood	4457 3129	590.5 ± 1.6 566.1 ± 3.1	590.5, 625	566.1 ± 3.1
B7	*	Z1410	Bentonite	Stretton Shale Formation	Bridleways Hazler Hill	4579 9335	566.6 ± 2.9 552 526	1500	566.6 ± 2.9
Lo10	10893	95-383	Lapilli tuff	Lightspout Formation	Lightspout Waterfall Longmynd	4303 9506	582.7 ± 3 555.9 ± 3.5 532.4 ± 3.8	582.7, 603, 646	555.9 ± 3.5
Arv2	10894	Z2237 95-141	Ignimbrite	Padarn Tuff Formation	Near Bryn Bras Castle	2553 3619	604.7 ± 1.6	617, 638	604.7 ± 1.6
Arv6	10895	Z2237 95-144	Ignimbrite	Padarn Tuff Formation	Track East of Bryn Bras	2561 3610	605.9 ± 3.8		605.9 ± 3.8
Arv7	10897	95-145	Tuff	Fachwen Formation	A4086 Lake View Hotel	2567 3611	574.3 ± 2.7	603	574.3 ± 2.7
Arv4	10896	Z2254 95-142	Glassy tuffite	Fachwen Formation	Track East of Bryn Bras	2564 3609	572.5 ± 1.2	620, 657	572.5 ± 1.2

*No sample remains of this bentonite.

Charnwood Terrane

The Charnwood terrane contains two areas of outcrop, Nuneaton and Charnwood Forest. The Nuneaton sequence is relatively simple: a volcanic formation cut by two sets of intrusions, clearly overlain unconformably by the Lower Cambrian (Tommotian) Hartshill Quartzite. The later of the two plutonic suites (granophyric diorite or markfieldite) has been dated (U–Pb zircon) at 603 ± 2 Ma (Tucker & Pharaoh 1991). In Charnwood Forest the sequence is more complex but more extensive. The Charnian Supergroup is a bedded sequence of volcanic rocks (largely tuffs) with, at the top of the sequence, a thick pelite devoid of volcanic input. Occurrences of fossiliferous tuffs within the supergroup occur within the Beacon Hill and Bradgate Tuff Formations. *Charnia masoni* (Ford 1958) was the first fossil to be discovered and subsequently other soft-bodied fossils belonging to the Ediacara fauna have been found at various localities in Charnwood Forest (Ford 1980, 1994). On a global scale the Charnwood Forest faunas probably represent older Vendian faunas (Runnegar & Fedonkin 1992). This bedded sequence has been shown to be stratigraphically continuous, though with some fairly rapid lateral facies changes, over a strike length of more than 15 km around the Charnwood Forest Anticline (Moseley & Ford, 1985). The uppermost formation of the sequence has recently yielded trace fossils of very early Cambrian age (Bland 1994; Ford 1994; Bland & Goldring 1995) and the stratigraphy has thus been subject to a major revision (McIlroy *et al.* 1998). The uppermost group, the Brand Group, is separated from the Vendian by unconformities at two levels and the uppermost formation, at least, is now regarded as of Lower Cambrian age. Complexes of intrusive and supracrustal volcanic rocks that are of small lateral extent interrupt the volcanic sequence in the west of the area at High Sharpley, Charnwood Lodge, Whitwick and Bardon Hill. These are usually regarded as replacing the bedded sequence in those areas. Small intrusions of mafic diorite (the Northern Charnwood Diorites) and large intrusions of granophyric diorite (the Southern Charnwood Diorites or markfieldites) cut the bedded sequence (Fig. 2). They both precede a penetrative cleavage (Boulter & Yates 1987) which transects the Charnwood Forest Anticline. The chemical similarity of the Southern Charnwood Diorite with the 603 Ma granophyric diorite of Nuneaton has led to suggestions that the whole of the Vendian sequence at Charnwood is pre-603 Ma, although it has always been recognized that this is much older than ages determined for other Vendian rocks (Pharaoh & Gibbons 1994; Pharaoh & Carney 2000). We have thus sampled part of the bedded succession to try to resolve this anomaly and to date the type Charnian fauna.

The sample obtained from within the main sequence of the Charnian Supergroup (field number CH2) comes from a road-cutting on the south side of the A50 near Markfield ([44861 31095], Fig. 2). It is the coarse tuff matrix to a slump breccia which has large pink slivers of a rhyolite or fine rhyolitic tuff within it. Moseley & Ford (1985) map the horizon here as the top of the Beacon Hill Formation since the Sliding Stone Slump Breccia Member, a very distinctive horizon, crops out on the hillside just above this locality. The horizon sampled is described as one of only three exposures of the Park Breccia Member by Worssam & Old (1988, p. 13) which is the pull-apart member 98.3 m. below the top of the formation noted by Moseley & Ford (1985, p. 9). Both they and Old & Worssam (1982, also Worssam & Old 1988) show this horizon stratigraphically continuous from this sample locality, across the Newtown Linford Fault, to the same horizon outcropping in Bradgate Park. There *Charniodiscus* can be seen on the bedding planes of the Old John Member below the War Memorial Obelisk a few metres above the Park Breccia Member.

We also attempted to date the Bardon Hill Complex, as this seems to cut the volcanics and might therefore yield a younger

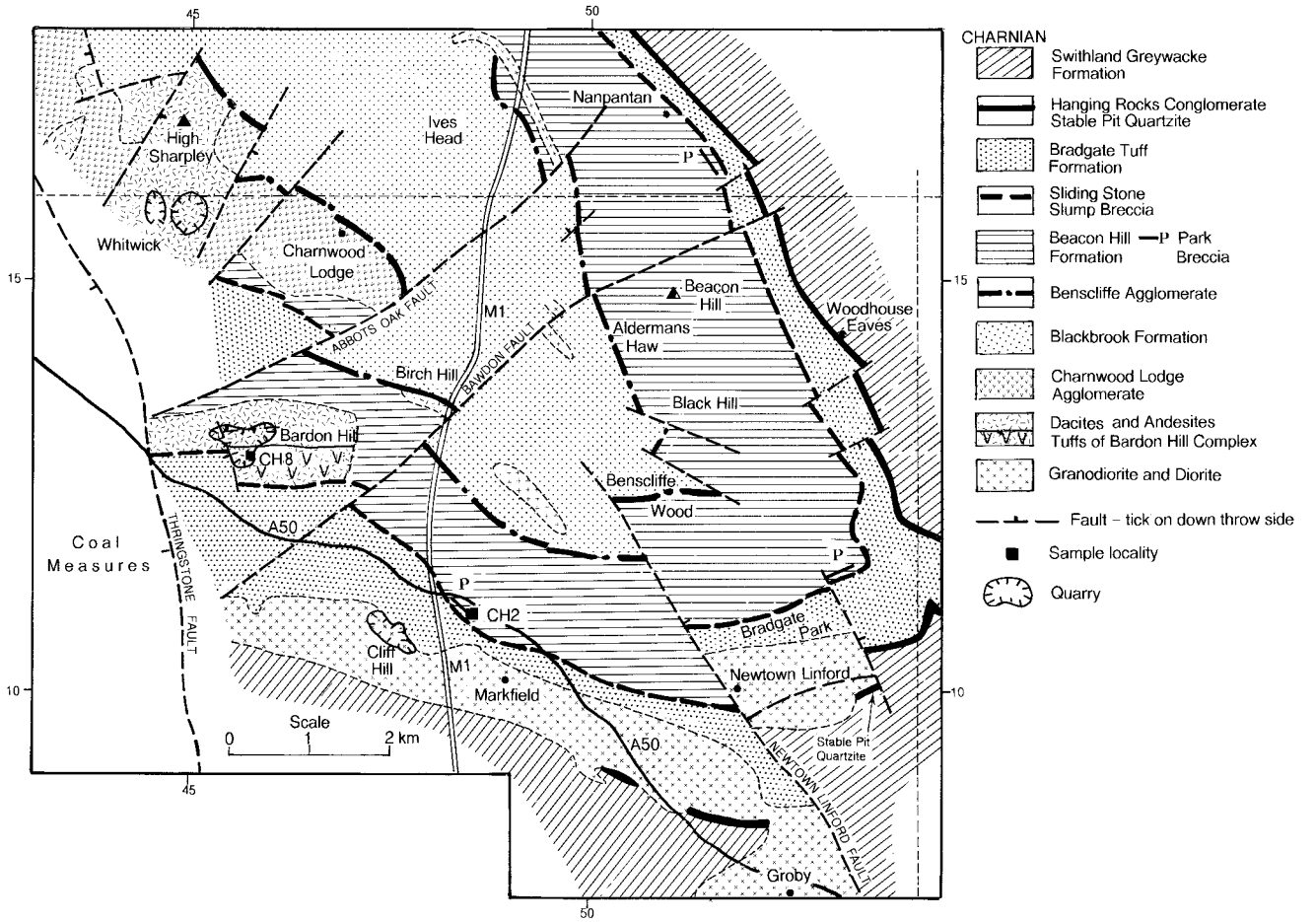


Fig. 2. Locality map of the Charnwood Forest area (adapted from Worssam & Old, 1988). Numbers at margin mark National Grid kilometre squares. CH2 & CH8 are sample localities.

age, but no zircons were obtained. However, we did obtain dateable material from the bedded volcanics lying to the south of the Bardon Hill Complex (CH8 at [4457 3129], Fig. 2). These volcanics are of different facies to the normal sequence of the Beacon Hill Formation and contain a major local detrital component derived from the Bardon Breccia, part of the Bardon Hill Complex (Carney & Pharaoh 2000*b*).

Specimen CH8 is a coarse leucocratic tuff containing fragments up to 1 cm. of rhyolitic and dark glassy volcanic rocks. The fragments are angular and the feldspar fragments fresh, although there are considerable amounts of chloritic alteration products. On the sample locality map (Fig. 2) we have attempted to distinguish between two components, the bedded volcanics occuring in the southern extension to the quarry and also in the land to the south and east, which has been extensively drilled by the Bardon Roadstone plc (D. Hopkins pers. comm.), and the more massive andesites and dacites forming the northern part of the quarry, which may have an intrusive component. Moseley & Ford (1985, p. 15) suggest that the Bardon Hill Complex is underlain by the Benscliffe Member, the lowest horizon of the Beacon Hill Formation, and both they and Old & Worssam (1982) show the complex as overlain by the Sliding Stone Slump Breccia Member, the lowest horizon of the next higher formation. Thus previous workers all ascribe these volcanics at Bardon Hill to the proximal deposits of a central volcano which lies stratigraphically within the Beacon Hill Formation and are thus lateral correlatives of the *Charniodiscus* bearing beds.

Wrekin Terrane

The Wrekin Terrane includes outcrops at the Malvern Hills and in the Welsh Borders (Wrekin, Stretton Hills, Long Mynd, Pontesford, Stanner–Hanter). A small outcrop at Llangynog containing the only other Ediacaran fauna found in Britain may also lie in this terrane. At both the Wrekin and Malvern there is unequivocal evidence of plutonic, volcanic and tectonic events before the Cambrian (Tommotian–Atdabanian) transgression. At the Wrekin the Ercall Granophyre (560 ± 1 Ma) cuts Uriconian Volcanics dated as 566 ± 2 Ma (Tucker & Pharaoh 1991) overlain unconformably by the Atdabanian Wrekin Quartzite (Cope & Gibbons 1987; Wright *et al.* 1993). In the Stretton Hills and on the Long Mynd there is a sequence of largely arenaceous beds, the Longmyndian Supergroup, whose age is unknown. It comprises a very thick (*c.* 7 km) sequence of largely clastic sediments with, at the base, many volcaniclastic and bentonitic horizons. It lies within the Church Stretton Fault System, being bounded to the NW by the Pontesford–Linley Fault. On the SE side the Church Stretton Fault forms much of the boundary but splays of this fault cut between the Longmyndian and the Uriconian, which crops out to the SE. Until recent years the Longmyndian has

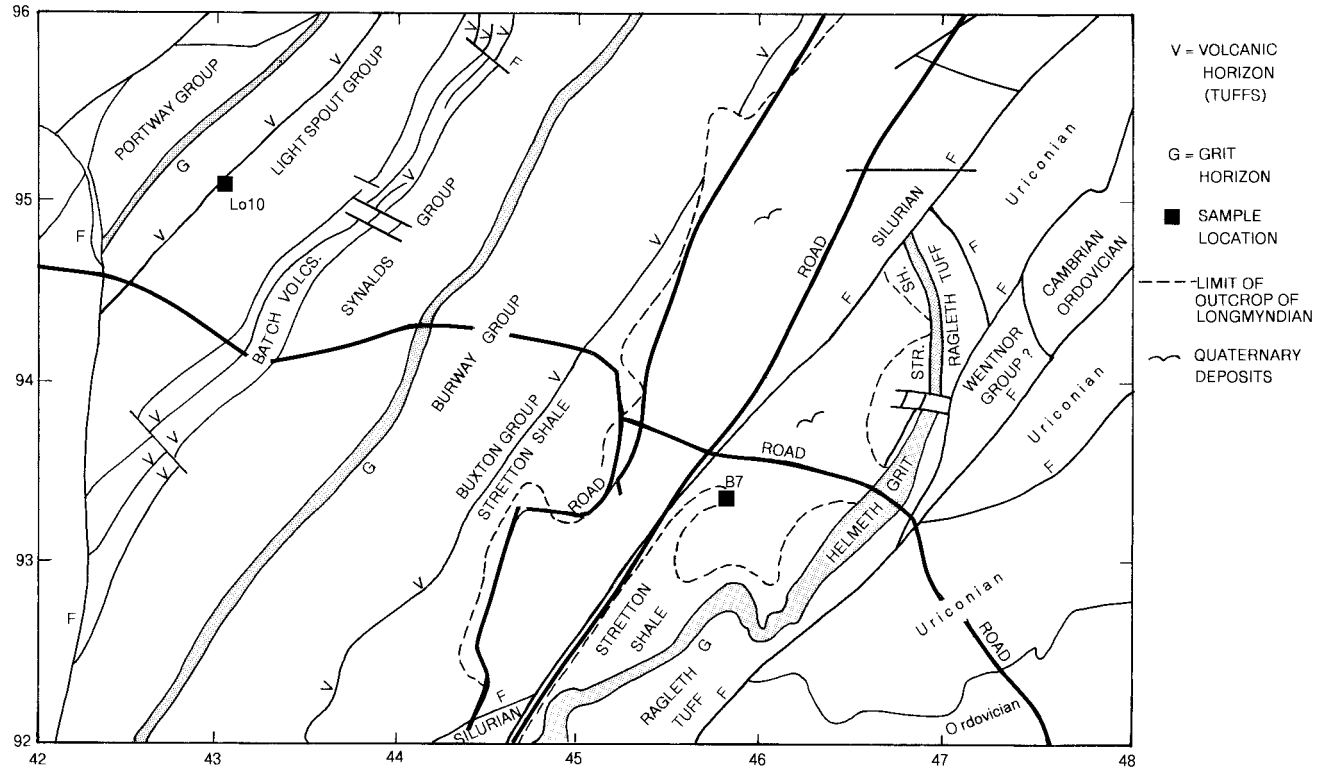


Fig. 3. Locality map of the Longmynd area (adapted from Dunham 1967). Numbers at margin mark National Grid kilometre squares. B7 & Lo10 are sample localities.

generally been regarded as a younger group than the Uriconian. Wilson (2000) quotes James (1952, 1956) to suggest that it is firmly established that Uriconian rocks underlie the Longmyndian east and west of the Long Mynd. However it has long been recognized (Wright 1969) that the western outcrops of the Uriconian are more convincingly interpreted as thrust over the Longmyndian and to the east the Ragleth Tuff, previously included in the Uriconian, is put by Pauley (1990) as the lowest formation of the Longmyndian. There is thus no direct evidence of the relative age of the Uriconian and the Longmyndian as all contacts are faulted. The Stretton Group is unconformably overlain by the Wentnor Group, which is entirely clastic, inviting correlation with the upper part of the Charnian Supergroup.

Several bentonitic clay horizons within the Stretton Group have been exposed by temporary excavations in recent years, all on the east side of the Church Stretton Fault. One horizon stratigraphically just above the Helmeth Grit was sampled by P.T. and dated by the fission-track method (526 ± 28 Ma, Naeser *et al.* 1982). Further temporary exposures at Hazler Orchard and Bridleways (two new housing developments) in this same area were sampled by all three authors. The sample B7 yielded abundant zircons but no useful material was obtained from the other two samples. B7 was taken from the road Bridleways, newly constructed in 1990 to the east of the Church Stretton valley on the slope of Hazler Hill, ([4579 9335], Fig. 3). It is from a *c.* 10 cm. thick bentonitic grey clay horizon within the monotonous shales and thin greywackes of this part of the sequence, about 300 m. above the base of the Stretton Shale.

Samples of the volcanic horizons within the upper part of the Stretton Group to the west of the Church Stretton Fault were collected with a view to determining the time span of the deposition of this thick clastic sequence. James (1956) and Pauley (1990) have mapped several volcanic horizons. The highest are those at the Lightspout waterfall, well up in the Lightspout Formation ([4303 9506], Fig. 3) and about 50 m. below the mapped boundary of the Portway Formation (the highest formation in the Stretton Group). Sample Lo 10 is a coarse lapilli tuff lying immediately above a fine white bedded tuff. This horizon is probably about 4 km. thickness above the base of the Longmyndian, rather more than half the total thickness of the Supergroup. (The thickness estimates include the *c.* 1 km. Ragleth Tuff as the basal formation of the Supergroup giving a total thickness of *c.* 7 km. for the whole.)

Cymru Terrane

The Cymru Terrane comprises four disparate areas: the Johnston Complex and St Davids area (Pebidian Supergroup) of South Wales, the Padarn and Bangor ridges (Arfon Group) and the Sarn Complex of North Wales. We have not sampled the South Wales areas, which have very different Precambrian–Cambrian sequences to those in North Wales being similar to the sequences in the Welsh Border. Only imprecise age data are available from South Wales, the Johnston *c.* 650 Ma and the Pebidian >570 Ma (Patchett & Jocelyn 1979), with the Pebidian deformed at greenschist facies before the Early Cambrian marine transgression. This is markedly different from the situation in North Wales. Here there are thick sequences of Precambrian and Lower Cambrian sediments. The Precambrian Arfon Group is made up of two distinct formations separated by an unconformity (Reedman *et al.* 1984; Pharaoh & Gibbons 1994). The lower, Padarn Tuff Formation, is everywhere a series of ignimbrites, but the upper series of formations differ in the different localities. In the

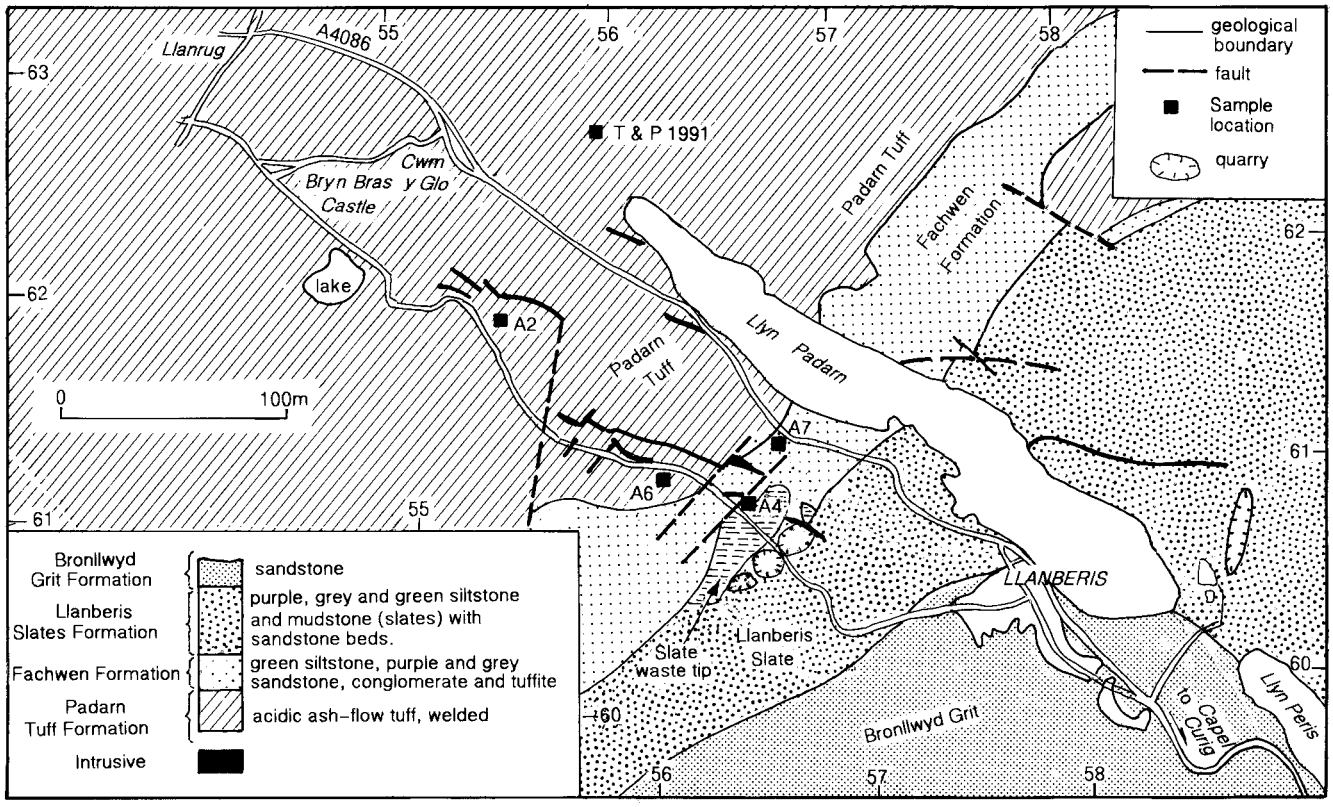


Fig. 4. Locality map of the Llanberis area (adapted from Howells *et al.* 1981 and Roberts 1979). Numbers at margin mark National Grid kilometre squares. A2, A4, A6 & A7 are Arv sample localities, T&P 1991 is the sample locality of ignimbrite dated by Tucker & Pharaoh (1991).

Llanberis area the upper formation is the Fachwen, a sequence of conglomerate and sandstones with tuffs, passing up into argillaceous slate with a few fine tuffs. The slate continues without apparent stratigraphic break for about 5000 m, unfossiliferous except for *Teichichnus* trails in the upper part, before slates which contain Lower Cambrian fossils. An ignimbrite from the Padarn Tuff Formation has been dated at 614 ± 2 Ma (Tucker & Pharaoh 1991) but the Fachwen tuffs are undated. As they are regarded as in stratigraphic continuity with the Llanberis Slate it was thought that they could yield an age close to the Precambrian–Cambrian boundary. Another point of interest is that correlations have been tentatively suggested between the Fachwen and outcrops of otherwise unknown affinity that lie unconformably on the deformed Precambrian rocks of Anglesey, in the Monian Composite Terrane (Reedman *et al.* 1984; Gibbons 1990). The Trefdraeth Conglomerate, Carreg Onen Beds, Baron Hill Beds and the Bwlch Gwyn Tuff lie on at least two different terranes into which the Mona Complex has been sliced by transcurrent faults. Any correlation across the Menai Strait Fault System would be of great significance.

Within the Menai Strait Fault System lies the fourth outcrop assigned to this terrane, the Sarn Complex. It has a deformed gneissose rock, the Parwyd Gneiss, and a granite dated at 615 ± 2 Ma (Horák *et al.* 1996). Because this is of similar age to the Padarn ignimbrite this sliver has been placed in the Cymru Terrane (Pharaoh & Carney 2000).

The Padarn Tuff is made up entirely of ignimbrite flows, several of which were sampled and one, Arv 2 from near the start of the track from Bryn Bras Castle to the slate quarries ([2553 3619], Fig. 4), gave abundant idiomorphic zircons. Zircons from a further sample (Arv 6, [25610 36105]) at a position near the top of the Padarn Tuff were mainly large idiomorphic laths.

The contact between the Padarn Tuff Formation and the Fachwen Formation in this vicinity is faulted but nearby can be shown to be an unconformity, the Padarn Tuff having been tilted before the deposition of the conglomerate at the base of the Fachwen (Reedman *et al.* 1984). Although the Fachwen is regarded as conformable with the Cambrian Llanberis Slate, in the area where the samples were taken there is a large gap in exposure (waste slate from the old slate quarries), so it is not certain that there is stratigraphic continuity between our sampled horizon and the Llanberis Slate.

The highest tuff bed in the exposed sequence of the Fachwen (a very thin glassy tuffite, Arv 4) crops out by the track just before the waste slate ([25635 36095], Fig. 4) (Howells *et al.* 1981). It is interbedded with argillaceous slate. Other tuff horizons occur lower in the Fachwen. The tuff sample, Arv 7, was taken from the lower of the two horizons cropping out beside the A4086 80 m. south of Lake View Hotel ([2567 3611], Fig. 4) (Roberts 1979), where a sequence of sediments can be seen, starting with a conglomerate, although the actual unconformity onto the Padarn Tuff is not visible here.

Analytical methods and interpretation of ages

Basic analytical methods for SHRIMP ages (Compston *et al.* 1984) can be obtained from the Society Library or the British Library Document Supply Centre, Boston Spa, Wetherby, West Yorkshire LS23 7BQ, UK as Supplementary Publication No. SUP 18169 (26 pages). Compston (2000) described recent procedures for data assessment that allow for Pb loss within the SL13 reference zircon and for

testing the sensitivity of the calculated ages to the instrumental discrimination gradient (slope of ln Pb/U v. ln UO/U). These procedures and checks have been followed here for each analytical session and results recorded in the Supplementary Publication.

The age distribution is displayed here using the kerned probability-density diagram (Silverman 1986), which is a form of histogram that allows for the different precision of the individual ages (caused by their differing U contents or differing instrumental stability), and whose shape does not depend on the choice of bin-width. This type of diagram correctly shows the distribution of measured ages, and peaks in the distribution may indicate the presence of different age-components such as older detrital grains or younger zircon overgrowths. However, interpretation of the plot is limited by analytical error. The distribution of *true* ages may be only approximated by the measured distribution, because the latter can be disturbed by fluctuations in measured age caused by instrumental error. Results of numerical simulations using known ages and random synthetic errors show that a continuous sequence of ages such as that produced by variable Pb loss will generally appear as two or more peaks rather than a plateau. This is a consequence of the sizes of the errors relative to the width of the plateau: as precision is made progressively higher, the number of peaks increases and their definition decreases until the plot approximates the required plateau. For a truly continuous age distribution, only the oldest peak has geological meaning and this is the age of the undisturbed zircons (assuming that at least some spots have not lost Pb). On the other hand, for a mixture of two or more discrete ages rather than a continuum, the probability density plot correctly shows separate real peaks as long as their age difference is big enough relative to analytical error.

Mixture-modelling (Sambridge & Compston 1994) enumerates the same peaks seen qualitatively in probability-density plots, by way of an independent mathematical method and gives precision estimates for the peaks that are objectively related to the precision of the individual analyses.

Tables of analytical results for the sample zircons are given in the Supplementary Publication. Uncertainties given in the Tables and text as ' ± ' are the standard error unless explicitly stated to be the 95% confidence limits.

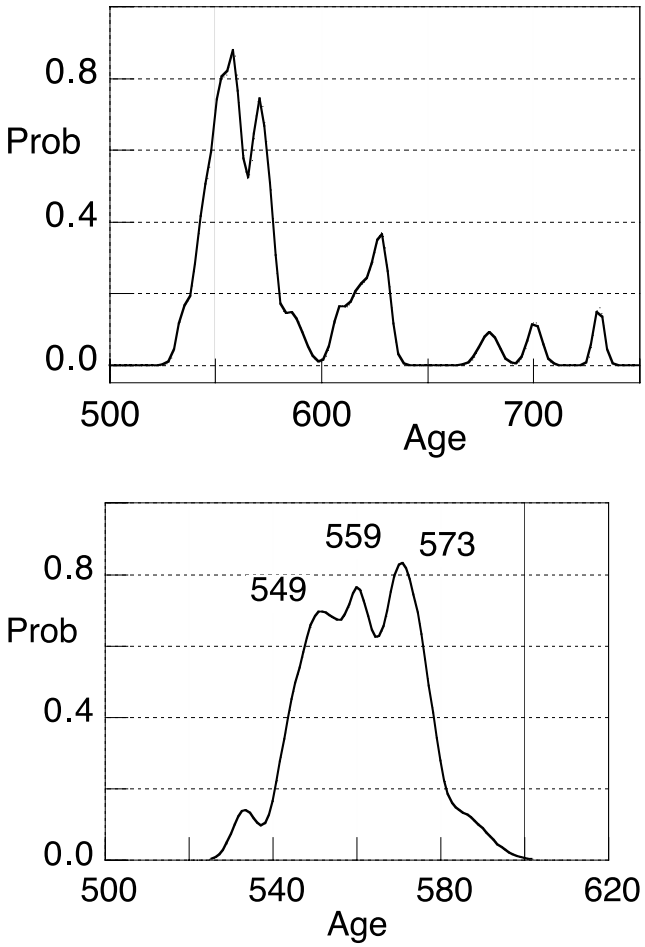


Fig. 5. (**a**) CH2 probability-density for ages <730 Ma. (**b**) CH2 probability-density for ages <600 Ma.

Charnian Supergroup age interpretations

Sample CH2, tuff matrix, Park Breccia Member, Markfield road-cutting

CH2 yielded a large number of fine-grained zircons (<50 μm). Most grains are idiomorphic but broken. A few were obviously detrital having rounded and pitted surfaces and none of these was analysed. Cathodoluminescence imaging (CL) showed that all grains contained euhedral internal zoning, some contained ovoid zircon cores, and that discernible zircon overgrowths were absent.

The sample ages for each separate session indicates the presence of three similar age-groups. The preferred age estimates come from the combined ages for both sessions. The distribution of the CH2 ages are shown by the probability density plot, Figure 5a, which emphasizes the presence of inherited grains/cores within the zircons. There are individual near-concordant ages at *c.* 1015 Ma and *c.* 1950 Ma. More importantly, there are composite but definite groups between 600 Ma and 640 Ma and older individuals ages up to 750 Ma, so it is quite clear that the tuff contains many inherited grains. The ages younger than 600 Ma can be resolved visually into three age-groups in the probability plot Figure 5b which is drawn at higher resolution. Mixture modelling enumerates these three groups quantitatively as 573.2 ± 1.0, 559.3 ± 2.0 Ma and 548.7 ± 1.9 Ma.

The interpretation problem is to know which of the younger peaks is the tuff magmatic age. The conservative interpretation, consistent with the discussion above of the use of probability-density diagrams, is to take the 573 Ma peak as the tuff age and both younger 'peaks' as artefacts in a continuous Pb loss distribution. An alternative is to take the tuff age to be the 559 Ma, with the 573 Ma peak being an inherited age-group and the younger peak being an artefact.

The question of Pb loss can be examined through the replicate analyses made in different areas within a number of the CH2 grains. The filled symbols in Figure 6 are replicates that agree within errors, the open symbols are those that either do not agree or are single analyses. The presence of close replicates at *c.* 575 Ma in two grains and at *c.* 585 Ma in another suggests that discrete grains of those two ages are present, and likewise two grains at *c.* 560 Ma. The latter gives some confidence that the 559 Ma age-group can be interpreted as the tuff age, and the 573 Ma group and all ages older than 600 Ma would then be inherited grains. The grains having duplicates <550 Ma are more complicated: they all contain at least one age that has been excluded as an older outlier, and the 549 Ma age-group can be viewed therefore as produced by Pb loss.

Sample CH8, tuff, Bardon Hill Complex, Bardon Hill Quarry

The 14 zircons from CH8 are mainly euhedral but broken, but the smallest grains (<30 μm) are rounded and smooth perhaps

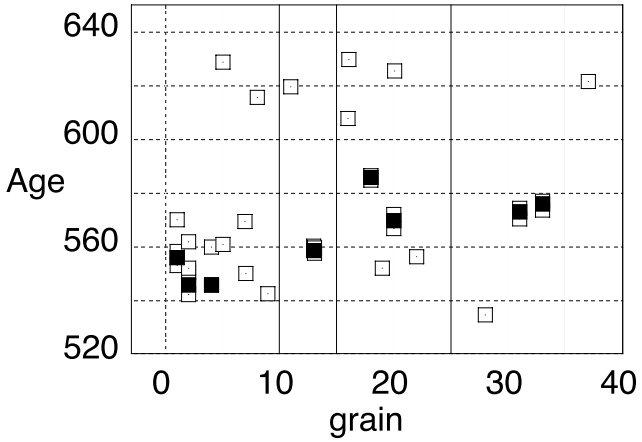


Fig. 6. CH2 replicate and single ages <640 Ma.

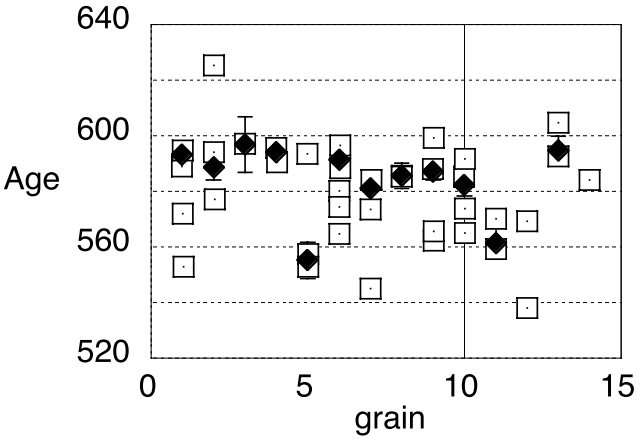


Fig. 8. CH8 ages, grain means and outliers.

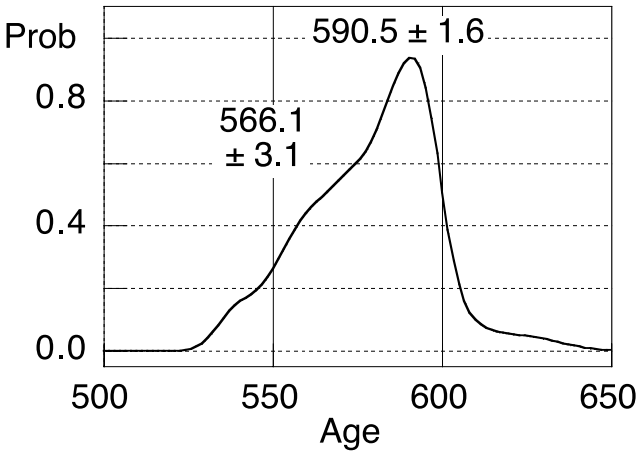


Fig. 7. Probability-density all CH8 ages.

indicating resorption. All contain needles or rods of apatite and some contain sulfide inclusions. No zircon cores were seen in transmitted or reflected light, but cores could be present as CL imaging was not done.

Ages were combined from two analytical sessions (Supplementary Publication) to form the probability plot Figure 7, and possible age components were enumerated using mixture modelling.

The age of the main group is well defined at 590.5 ± 1.6 Ma (21 analyses). A possible younger component at 566.1 ± 3.1 Ma (13 analyses) is indicated by the inflexion and strong asymmetry of the Figure 7 plot, and there are two isolated ages at 542 ± 6 Ma and 625 ± 11 Ma. The question now is the reality or otherwise of the 566 Ma age, which is within error of the 559.3 Ma age for the Beacon Hill tuff (sample CH2) considered to be at the same stratigraphic level. It may signify zircon crystallization at that time, in which case it would represent the age of volcanism and the main 590 Ma age group would be interpreted as inherited. Alternatively, it may represent an artefact formed by areas within grains that have lost similar fractions of their radiogenic Pb.

Figure 8 presents the weighted mean values for two or more ages within grains that agree with each other (filled diamonds), and single or two ages (open squares) that disagree with others within the same grain. For ten grains, the (selected) replicate ages agree at 588.7 ± 1.3 Ma and for two, grains 5 and 11, they agree at *c.* 559 Ma, of which grain 5 has a *c.* 594 Ma outlier also. Three of the ten grains show one or more low ages that range from 538 Ma to 574 Ma, and one has a high outlier at *c.* 625 Ma. These observations support the presence of 590 Ma old zircon grains in the tuff and the occurrence of sporadic Pb loss within those grains. If 590 Ma is interpreted as the age of tuff magmatism, it follows that the two age duplicates at 559 Ma were produced by coincidence in Pb loss in older grains.

$^{207}Pb/^{206}Pb$ ages and the Tera–Wasserburg Concordia presentation (Fig. 9a, b) were used in an attempt to confirm the 590 Ma and 566 zircon ages as separate. The CH8 analyses were favourable for measurement of the radiogenic $^{207}Pb/^{206}Pb$ because the grain-mount was particularly low in surface Pb contamination. A fixed blank was therefore subtracted from the total ^{207}Pb and ^{206}Pb for each spot per session rather than its individual common Pb estimate, which is usually much less precise.

The open squares in Figure 9a represent the total ^{206}Pb data and the filled diamonds the radiogenic. The mean error bars are shown at the point 0.072, 11.0, and Concordia is marked at 10 Ma intervals. One of the radiogenic points is a single low outlier below Concordia, and three that lie obviously above it can be interpreted as older grains that have lost Pb (or spots that contain more common Pb than assumed). Three other points are detectably above Concordia. Figure 9b identifies grains 5 and 11 which give duplicate $^{206}Pb/^{238}U$ ages at *c.* 560 Ma, and which are within error of Concordia. This is permissive evidence that their undisturbed age is indeed *c.* 560 Ma, although their $^{207}Pb/^{206}Pb$ ages alone are not sufficiently precise to prove it.

The $^{207}Pb/^{206}Pb$ ages for the remaining 31 analyses behave as a single population (MSWD 1.25), equivalent to a weighted mean $^{207}Pb/^{206}Pb$ age of 585 Ma (Fig. 10). No younger population at *c.* 566 Ma can be detected using mixture-modelling as all the scatter is consistent with the analytical errors.

Simulations for 31 $^{207}Pb/^{206}Pb$ ages using random blends of 590 Ma and 566 Ma ages in the proportion 4:1 show that the error per age must be *c.* 20 Ma or less before the MSWD becomes significant and mixture-modelling can resolve the two ages. Although the Concordia approach correctly gives the age of the major component, it cannot in this case test the hypothesis that the 566 Ma age indicated by the $^{206}Pb/^{238}U$ ages is also present.

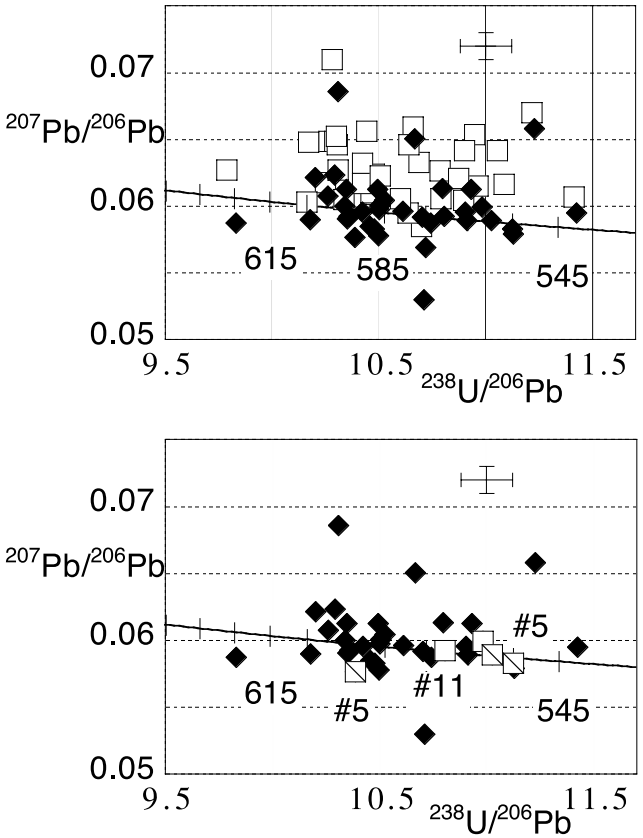


Fig. 9. (**a**) CH8 Concordia plot. (**b**) CH8 Concordia plot, grains 5 and 11 distinguished.

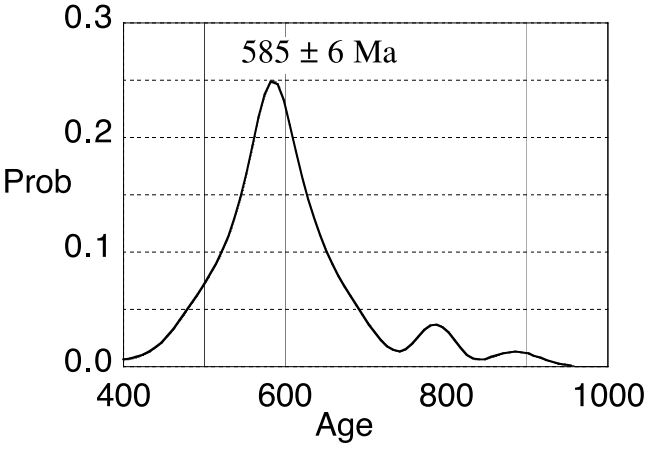


Fig. 10. CH8 $^{207}Pb/^{206}Pb$ ages, probability-density.

Longmyndian Supergroup age interpretations

Sample B7, bentonite, Bridleways, Church Stretton, Longmyndian

The zircons obtained from this sample are of mainly two morphologies: laths of *c.* 50 μm width and having a length/breadth ratio of 4 or more, and short biprisms with a length/breadth ratio less than 2 and of similar width. Most laths are broken and both forms are idiomorphic. A few small rounded and pitted grains were also found.

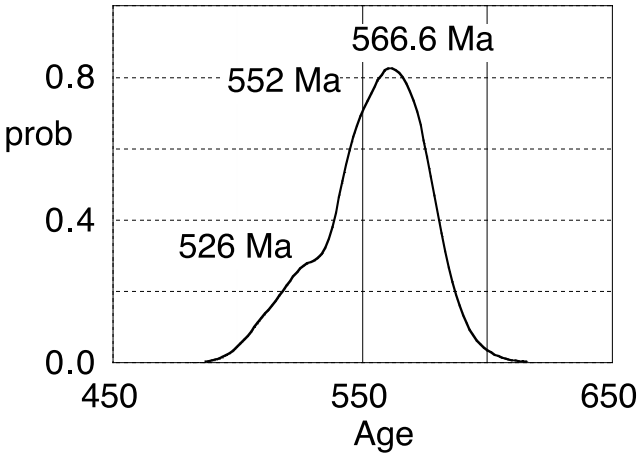


Fig. 11. B7, age probability-density.

Ages from the two analytical sessions were combined to form the probability-density plot, Figure 11. Because the oldest age-group at 566.6 ± 2.9 Ma is much the largest and because zircon laths are commonly associated with rapid crystallization, it most likely denotes the age of tuff magmatism. The two younger 'groups' are taken as artefacts of small amounts of Pb loss as seen by the mixture-modelling operation. No correlation was found between age and the two main zircon morphologies. Two grains (#41,42 in Supplementary Publication) of age *c.* 1500 Ma are either detrital without sign of abrasion or, more likely, xenocrystic.

Sample Lo 10, tuff, Lightspout Formation, Lightspout waterfall, Cardingmill Valley

A total of 14 small zircons, 50–100 μm, were obtained from this sample, mostly broken fragments of originally larger prismatic grains. BSE images made after analysis showed that the external surfaces of half the grains, including fracture surfaces, were prominently embayed. This implies magmatic resorption and suggests that all grains might be older than the tuff magmatism, although not necessarily much older. Most grains contained internal cracks which could not be avoided in the analysed areas, and a few had mineral inclusions. All grains showed euhedral zoning but only one had a discernible zircon core. The reference zircon used for Lo 10 was Arv 4, whose age had been determined earlier at 572.5 ± 1.2 Ma using the SL13 standard. The small Lo 10 grains were repolished and reanalysed to obtain 28 analyses in four sessions.

The combined age distribution in Figure 12 indicates the effects of variable inheritance within the ages, and possibly of Pb loss and instrumental factors as well. There is no dominant age mode but two separate modes of roughly equal abundance, so the first question is whether either of these two denotes the age of tuff magmatism.

In principle, the youngest group at 532 Ma (Fig. 12) may be either the tuff age or a group formed from areas that have lost small fractions of their Pb. Its interpretation as the tuff age is not excluded by any presently published ages from elsewhere. In particular, it is permitted by the age of 538.2 ± 2.0 Ma for the early Tommotian Bed 5 at Meishucun, South China (Jenkins *et al.* 2002). The 557 Ma group may be either an inherited age or the tuff age, depending on the interpretation of the 532 Ma group.

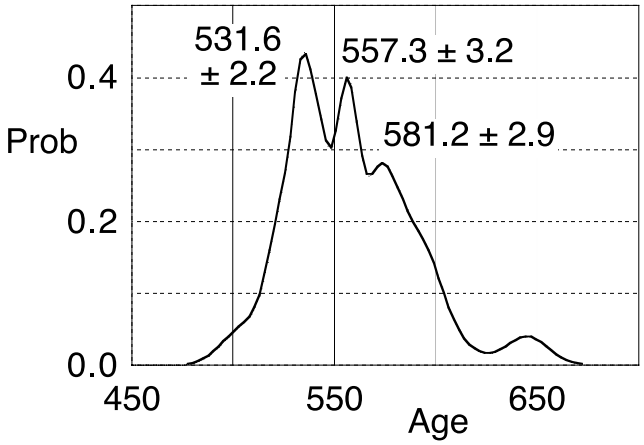


Fig. 12. Lo 10, age probability-density, combined ages.

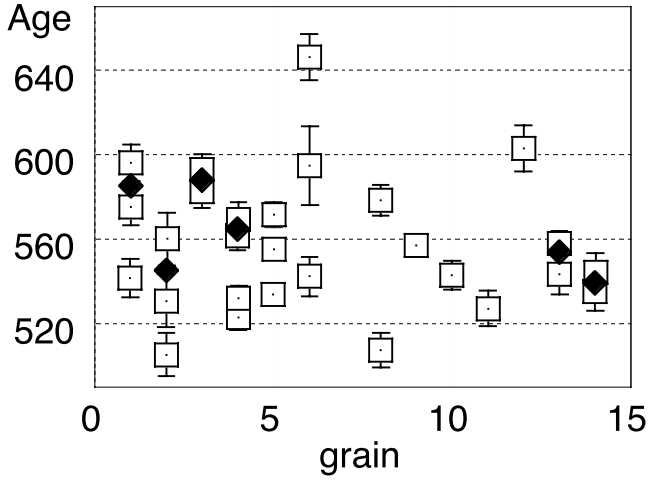


Fig. 13. Lo 10, age means and outliers per grain.

The second approach to interpreting the Figure 12 distribution is assessment of the replicate ages per grain, which is shown by Figure 13. Duplicate ages within four grains agree with each other (filled diamonds) as do selected pairs of ages within two others that were analysed three and four times. Ages within other grains such as #5 and #6 signal the presence of inheritance and/or Pb loss. The BSE images for the grains show a definite zircon core only in #9 whose age is in the middle of the range: this core was either not sampled by the probe or not significantly older than the overlying zircon. The lack of cores in the other grains requires that Pb loss is the process that has dispersed their ages, not internal inheritance.

It also follows that there are differences in age between grains, which is consistent with the impression of resorption given by the grain morphology: possibly all grains predate the final tuff volcanism. Grain 6 is older than any of the others, and it is probable that grains 1 and 3 are older than grains 2, 4 and 5. The (selected) duplicates in grains 2,4, 5 and 13 agree with each other within error at 558.9 ± 2.4 Ma, but the duplicate age for grain 14 is younger at 539.5 ± 6.3 Ma. The former matches the mixture-modelling result of 555.9 ± 3.5 Ma based on the ungrouped data-set, including grains having single analyses. The latter matches the 535 Ma age-group. The nub of the interpretation is whether the grain 14 duplicate age denotes a real time of zircon crystallization or whether instead it is the chance coincidence of two Pb-loss numbers. We are inclined to the latter alternative but the data do not allow us to exclude the former. In any case, we are not willing to base the time of volcanism on the age of a single zircon grain.

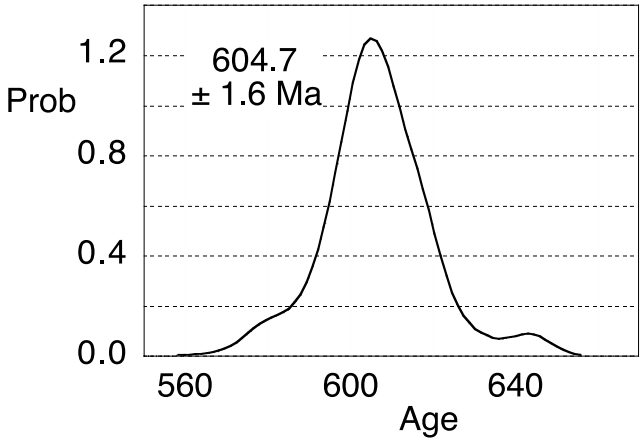


Fig. 14. Arv 2, age probability-density, combined ages.

Arfon Group age interpretations

Sample Arv 2, ignimbrite, Padarn Formation, Llanberis

This sample gave abundant idiomorphic zircons ranging in form from equant to broken and unbroken lath shaped grains, from one hundred to several hundred microns in length, and with inclusions of apatite needles and tubes of devitrified glass.

The combined ages from two analytical sessions show a single age group (Fig. 14), plus one younger age and two individual older ages within the same grain. (The latter grain does not differ visually from the others). We regard the predominant 605 Ma age group as the age of tuff magmatism.

Figure 15 shows the individual ages as open squares, and the weighted means for duplicates that agree within errors per grain as filled diamonds with error bars. Only the duplicates for grain 18 do not agree, and the mean for grain 29 is obviously greater than the remainder. Statistical analysis shows that the mean for grain 31 also exceeds the others, which latter agree to within errors at 604.7 ± 1.6 Ma.

The Arv 2 results show the precision with which zircon grains can be measured by SIMS when the target zircons have uniform Pb/U. Only one analysis shows Pb loss, two individual older grains have been detected, and the standard deviation per spot age as observed is ± 5.1 Ma in close agreement with the value expected from known ion counting errors. The Figure 15 duplicates contrast with most of the those measured for other zircons that have been disturbed by Pb loss.

Sample Arv 4, tuff, Fachwen Formation, Llanberis

Arv 4 contains clear idiomorphic zircons, 19 of which were analysed. CL imaging shows that no large and obvious zircon cores are present, and that the grains are euhedrally zoned throughout. Small cores may be present at the centres of a few grains such as the grain 2.04 analysis.

The largest age-group for the combined ages from two analytical sessions is the youngest (Fig. 16). Two groups of inherited grains are indicated, in addition to the two isolated grains older than 650 Ma.

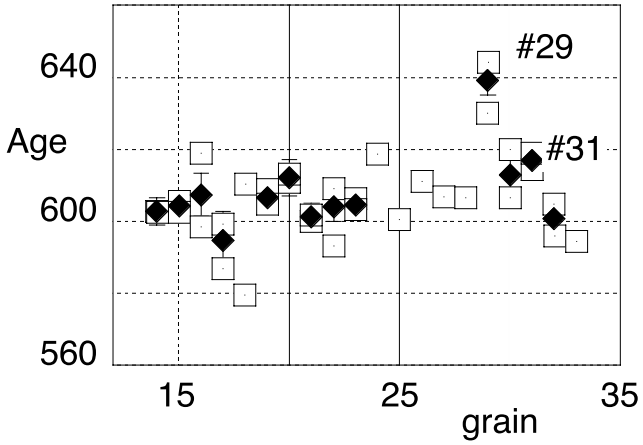


Fig. 15. Arv 2, age means and outliers per grain.

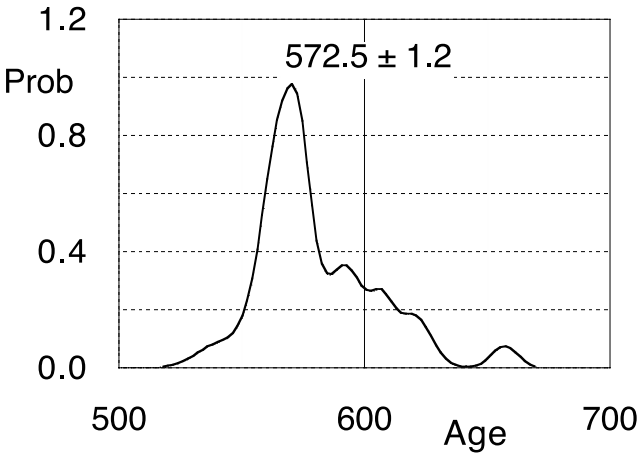


Fig. 16. Arv 4, age probability-density.

Different spots on nine grains were analysed to check for Pb loss and for discrete inherited grains. The single low age within grain 17 (Fig. 17) indicates Pb loss within the grain, while a single older age within grain 2 (657 Ma, beyond the range of Fig. 17) indicates inheritance within a discrete 600 Ma grain. Interpretations of the ages within grains 11, 12 and 13 are ambiguous. The three grains might be *c.* 570 Ma with inheritance, or they might be *c.* 620 Ma with Pb loss that lowers the ages to agree coincidentally with the younger age-group.

The age for the tuff volcanism is interpreted as 572.5 ± 1.2 Ma, corresponding to the largest age-group. The remaining analyses can be represented as a single population at 603 ± 3 Ma, but this would be a population of xenocrystic cores within the magmatic grains.

Sample Arv 6, ignimbrite, Padarn Formation and sample Arv 7, tuff,Fachwen Formation, Llanberis

Samples Arv 6 and Arv 7 were mounted on the same grain-mount as Arv 2 and only a small number of analyses were made of each, so it is convenient to assess their ages jointly despite their differing stratigraphic positions. The small numbers of analyses also mean that only tentative interpretations can be made. The Arv 6 zircon ages are mainly a *c.* 605 Ma group with one age having a large error at *c.* 580 Ma, whereas

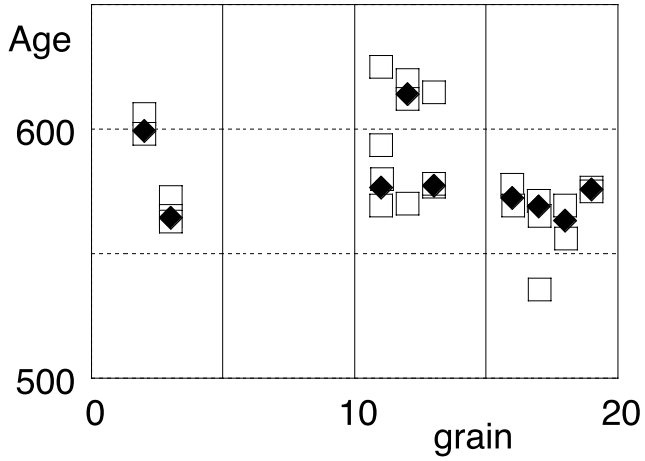


Fig. 17. Arv 4, replicate ages per grain.

Arv 7 is mainly *c.* 575 Ma with one age at *c.* 605 Ma. Although the 605.9 ± 3.8 Ma group age for Arv 6 nominally exceeds that for the Arv 2 Padarn Formation sample which is older on field evidence, the combined age uncertainties allow Arv 6 to be up to 7 Ma younger (95% limit). Because the Fachwen Formation is younger than the Padarn, the 574.3 ± 2.7 Ma age from Arv 7 probably denotes Fachwen Formation volcanism and the one 603 Ma age in Arv 7 would be an inherited grain or core. Of the two Fachwen samples Arv7, which is stratigraphically older than Arv4, does give a slightly older age, but within error is indistinguishable in age from the younger, Arv4, tuff.

Geological implications of the $^{206}Pb/^{238}U$ ages

Charnian Supergroup

The tuff dated from within the Beacon Hill Formation (CH2) can be accurately correlated with beds immediately below the *Charniodiscus*-bearing Old John Member. The interpreted age of 559.3 ± 2.0 Ma is comparable with that of other tuffs bearing a Vendian fauna elsewhere in the world. Martin *et al.* (2000) report 555.3 ± 0.3 Ma for zircons from tuff within a *Charnia*-bearing horizon at Zimnie Gory, North Russia, while Dunning (*in* Benus 1988) obtained 565 ± 1.5 Ma for a tuff from Mistaken Point in Newfoundland associated with an Ediacara-like fauna. Tuff in the Slawatycze Formation of Poland, regarded as the close of the early Vendian, gave an age of 551 ± 2 Ma (Compston *et al.* 1995). Although comparable, these ages are detectably different from each other and register a time-span of *c.* 15 Ma for the early Vendian. Tuffs in later Ediacaran assemblages in Namibia give $^{206}Pb/^{238}U$ ages in the range 540 to 544 Ma (Grotzinger *et al.* 1995), and $\leqslant$541 ± 2 Ma near the Vendian–Cambrian boundary in the Ukraine (Compston & Jenkins 1994). This confirms the contemporaneity of Ediacaran faunas worldwide.

The age of the tuff collected from bedded volcanic rocks newly exposed in the quarry at Bardon Hill (Fig. 2) is less easy to interpret. This is partly because the geological position of the sample is equivocal. The quarry lies within a sub-volcanic pluton and the quarrying now exposes a sequence, several hundred metres thick, of tuffs and agglomerates into which the pluton is intruded, though the sequence sampled is probably faulted against the Bardon Hill Complex (Carney & Pharaoh

Table 2. *Sequence of events in the Charnwood Forest and Nuneaton areas*

	Nuneaton	Charnwood Forest
Lower Cambrian	Purley Shale	
	Unconformity	
	Home Farm Member (Hyolithus Limestone)	
	Unconformity	
Tommotian	Hartshill Quartzite (lowest conglomerate contains boulders of Markfieldite)	
	Marine Transgression (unconformity on Caldecote Volcanic Fm)	
		Charnwood Forest Anticline and cleavage development
?Nemakit–Daldynian		Brand Group
		Swithland Formation
		Brand Hills Formation
		Unconformity
		Hanging Rocks Formation
		Unconformity
		Intrusion of Southern and Northern Charnwood Diorites
		Maplewell Group
Vendian faunas		Bradgate Tuff Formation
		Beacon Hill Formation; 559.3 ± 2.0 Ma[1]
		Bardon Hill Tuffs; 566.1 ± 3.1 Ma[1] (Inherited grains 590.5 ± 1.6 Ma)
		Blackbrook Group
	Granophyric Diorite (Markfieldite) 603 ± 2 Ma[2]	
	Mafic diorite sheets	
	Caldecote Volcanic Fomation	Morley Quarry older volcanics[3]

[1]This paper.
[2]Tucker & Pharaoh (1991).
[3]Pharaoh & Evans (1987).

2000*b*). The volcanic rocks of the Bardon Hill Complex and the tuff sampled are probably best interpreted as simply a thick proximal development of tuffs and agglomerates at a central volcano roughly comparable in age to the middle part of the Charnian (Moseley & Ford 1985). The majority of grains in the sample proved to be 590.5 ± 1.6 Ma but as there was also a population of ages, including several duplicate ages on individual grains, giving a mean of 566.1 ± 3.1 Ma we are inclined to interpret the data as indicating an eruption at *c.* 566 Ma in which the magmas or the pyroclasts include a preponderance of 591 Ma material. An alternative interpretation is that the 591 Ma date represents the time of eruption, with younger ages an artefact of lead-loss, and that the Bardon Hill Complex is an entirely fault bounded inlier of older volcanics. This would imply that the previous stratigraphic evidence of Moseley & Ford (1985) and Worssam & Old (1988) is an interpretation based on incomplete exposure.

These new age data and the revision of the stratigraphic age of the upper Charnian to a Cambrian age (McIlroy et al. 1998) makes a synthesis of the sedimentary, igneous and structural events still rather difficult. An age of 603 Ma for the Southern Charnwood Diorites (by petrochemical and petrographic correlation with the granophyric diorite of Nuneaton) becomes impossible to sustain as they can be seen to cut the Maplewell Group in the Cliffe Hill Quarry (Boulter & Yates 1987; Carney & Pharaoh 2000*a*) and the lower part of that group we have now dated as 559.3 ± 2.0 Ma. The Northern Diorites can also be seen to cut and contact metamorphose the Beacon Hill Formation (low in the Maplewell Group) (Boulter & Yates 1987) and thus must also be younger than 559.3 ± 2.0 Ma.

It is now apparent that in the Charnwood Forest and Nuneaton region there were two volcanic episodes developed after about 620 Ma followed by a clastic sedimentary sequence (Table 2). There is no stratigraphic or other geological evidence here for a break between the two volcanic episodes. The subdivision of the Charnian and Nuneaton volcanics into two age groups is not supported by their geochemistry which demonstrates that all the volcanics and the plutons are of the same rather distinctive type and they are regarded as belonging to one geochemical province (Pharaoh *et al.* 1987). The two hiatuses in the Upper Charnian recognized by McIlroy *et al.* (1998) occur above the Vendian volcanics, with the Cambrian rocks of the sequence being predominantly pelitic and with conglomerates above the upper unconformity containing diorite clasts in more than one locality. The revision of the Charnian stratigraphy to place the Swithland Formation in the Cambrian (McIlroy *et al.* 1998) is based on the occurrence of trace fossils, principally on the abundance of intensely churned *Teichichnus* ichnofabrics in the Swithland Formation. They compare the Charnian sequence with the well exposed sequences of late Precambrian and Cambrian age in SE Newfoundland where the Precambrian–Cambrian boundary stratotype has been designated. Here four cycles of sedimentation have been described (Landing 1992; Myrow & Hiscott

1993). The ascription by McIlroy *et al.* (1998) of the Hanging Rocks Formation of the Charnian to the top of the Vendian, which their detailed descriptions of 30–40 m sections at the golf course and Brand Hills show to be largely volcanogenic, seems entirely reasonable. We are not convinced, however, by their correlation of the Brand Hills and Swithland formations with the upper part of the Avalon sequence. They correlate the Stable Pit Quartzite of the Brand Hills Formation with the Hartshill Formation quartzites of Nuneaton. However, the Cambrian succession at Nuneaton barely 20 km to the SW is very different lithologically and palaeontologically as well as structurally. The Hartshill quartzites are 270 m thick with thick conglomerates at their base (Brasier 1989) whereas the Stable Pit Quartzite is only a few metres thick (although probably reduced in thickness by faulting) and from the evidence of their own sections (their fig. 3) does not extend as far as the Brand Hills. The Swithland Formation, a very thick mudrock universally turned into a deep purple slate, bears little lithological or palaeontological resemblance to the Home Farm Member and Purley shales of Nuneaton. Perhaps its closest analogue in the UK is the lower parts of the Llanberis slates of North Wales which also contain *Teichichnus*. We suggest that a more convincing correlation of the Swithland slate is with the Chapel Island Formation of Avalon placing it in the earliest Cambrian Stage, the Nemakit–Daldynian, which has been dated at 530.8 ± 0.5 Ma (Isachsen *et al*. 1994).

Further complications are made by the preliminary Acadian age reported by Carney (1999) for the cleavage of the Swithland Formation. This $^{40}Ar/^{39}Ar$ date on the micas is supported by the greenschist facies nature of the metamorphism based on illite crystallinity Kubler indices. Although this is, as he says, higher grade than the concealed Cambrian and Ordovician basement sampled in nearby deep boreholes, he omits to point out that at nearby Nuneaton the well-exposed Cambrian rocks are at much lower grade than the underlying Caldecote Volcanics. This is also the case for much of the Welsh borderland and South Wales, so the geological evidence is very strong for folding and cleavage formation at low greenschist facies of all these late Precambrian–earliest Cambrian sequences taking place before the Tommotian. We would thus propose that the folding and cleavage formation of the Charnwood Forest Anticline took place within the Nemakit–Daldynian and before the widespread Tommotian–Atdabanian marine transgression that took place over the English Midlands, Welsh borders and South Wales equivalent to the Random sequence of Avalon.

Longmyndian Supergroup

Dating the sequence in this area was thought to be crucial to any understanding of Late Precambrian events since the age of the Ercall Granophyre, which is overlain by Cambrian rocks containing Atdabanian fossils (Wright *et al.* 1993), gave an intrusion age of 560 ± 1 Ma (U–Pb zircon, Tucker & Pharaoh 1991) and a retrogressive event at 533 ± 12 Ma (Patchett *et al.* 1980). On the traditional view that the Longmyndian lies unconformably above the Uriconian, all the Longmyndian deposition and a major fold episode took place between 560 and 533 Ma. The only direct evidence of the age of the folding of the Longmyndian, both east and west of the Church Stretton Fault, shows it to be demonstrably pre-Llandoverian and possibly pre-Caradocian. East of the Church Stretton Fault, in the Cwms area, Cambrian Wrekin Quartzite rests unconformably on, or is in faulted contact with, possible Wentnor Group sediments (Greig *et al.* 1968, pp. 37 & 95) but the exposures in this area are very poor. The Uriconian at the Wrekin, however, is quite steeply dipping under the shallow Cambrian (Wright *et al.* 1993) and the folding and thrusting of the Uriconian in the Stretton Hills (Greig *et al.* 1968) can therefore confidently be regarded as pre-Atdabanian. A Stretton Shale bentonite was dated by the fission-track method (Naeser *et al.* 1982) and a date of 526 ± 28 Ma obtained. A similar date was obtained by the Rb/Sr method on shales from the Stretton Group (Bath 1974). These dates presumably register the age of uplift of this area after this fold episode (Pauley 1990; Wright *et al.* 1993).

Three bentonites were sampled by us and zircons from the sample near the base of the Stretton Shale Formation (B7) give an age of 566.6 ± 2.9 Ma. This is the same as a date obtained for the Uriconian Volcanics (566 ± 2 Ma, Tucker & Pharaoh 1991) and shows that Pauley (1990) was correct in suggesting that the two are laterally equivalent. Older grains within the sample yield dates of *c.* 1500 Ma. probably giving an indication of the age of the deep basement of this area of southern Britain. From a volcanic horizon close to the top of the Stretton Group we have obtained a date of 555.9 ± 3.5 Ma. It would thus seem that these rocks either side of the Church Stretton Fault are most likely to be part of one sequence (as has been traditionally thought) and that this fault does not represent a major terrane boundary.

We can thus be confident in confirming Pauley's (1990) hypothesis that the Longmyndian and the Uriconian are more or less contemporaneous. The sequence of events in this area is now much clearer (Table 3) with an extrusive volcanic episode *c.* 570–550 Ma gradually developing, in part of the area, into a clastic dominated trough, followed by a non-volcanogenic clastic extension of the basin. The age of a major fold episode, with thrusting eastwards within the Uriconian, may be recorded by the uplift (K/Ar) age of the Rushton Schist (Thorpe *et al.* 1984) and the Rb/Sr age of the Ercall Granophyre (Patchett *et al.* 1980) both about 535 Ma, close to the age of the base of the Cambrian. The fission-track and Rb/Sr ages on the Longmyndian bentonites and shales (Naeser *et al.* 1982; Bath 1974) also indicate uplift within the lowest Cambrian. There is therefore the possibility that this non-volcanogenic sequence and the fold episode are of very lowest Cambrian age (Nemakit–Daldynian). The transgression of the Cambrian sea in Tommotian–Atdabanian times cut a planar surface across this complex basement and deposited a sheet of Lower Cambrian sediments over the Uriconian, now seen at the Wrekin and at Comley, on the east side of the Stretton Hills.

Arfon Group

We have dated the ignimbrites of the Padarn Tuff formation at 604.7 ± 1.6 Ma and 605.9 ± 3.8 Ma. These are compatible with the date, 614 ± 2 Ma, obtained by Tucker & Pharaoh (1991) for an ignimbrite lower in the sequence. The date of 604.7 Ma we regard as the most accurate and least equivocal of the data presented here and the age of the Padarn Tuff can now be regarded as very secure. The sequence lying unconformably above the Padarn Tuff (the Fachwen Formation) is also now quite clearly dated as early Vendian since the highest tuff above the unconformity has given a date of 572.5 ± 1.2 Ma. The lower tuffs (still above the conglomerate) yielded only a few zircons which give a very comparable age of 574 Ma. As there has been no description of

Table 3. *Sequence of events in the Longmynd, Stretton Hills and Wrekin areas*

	Longmynd	Wrekin & Stretton Hills
	Comley Sandstone	Comley Sandstone
Atdabanian	Wrekin Quartzite	Wrekin Quartzite
	Marine Transgression	
	Unconformity on Uriconian and onto the Ercall	*Granophyre and ? on Wentnor Group*
	Uplift of Stretton Group c. 526 Ma[1,2]	
		Hydrothermal alteration and uplift of Ercall Granophyre 533 ± 12 Ma[3]
		Uplift of Rushton Schist 536 ± 8 Ma[4]
	Folding of Longmyndian into overturned fold	*Thrusting of Uriconian*
?Nemakit–Daldynian	Wentnor Group (finer clastics)	
	Unconformity overstepping onto Uriconian	
	Stretton Group(coarse clastics, tuffs and bentonites) 566 ± 2.9 Ma to 555.9 ± 3.5 Ma[6] (inherited zircons *c.* 1500 Ma)	Intrusion of Ercall Granophyre 560 ± 1 Ma[5]
		Uriconian Volcanism 566 ± 2 Ma[5]
		Deformation and Metamorphism (to form Rushton Schist) 667 ± 20 Ma[4]
		Deposition of Rushton Formation

[1]Naeser *et al.* (1982).
[2]Bath (1974).
[3]Patchett *et al.* (1980).
[4]Thorpe *et al.* (1984).
[5]Tucker & Pharaoh (1991).
[6]This paper.

Table 4. *Sequence of events in the Arfon area, North Wales*

		Llanberis Slates	Pelites with body fossils
Lower Cambrian			Pelites with *Teichichnus* trails Barren pelites
Vendian	Arfon Group	Fachwen Formation	Pelites with tuff 572 ± 1.2 Ma[1] (inherited zircons 620–657 Ma)[1] Tuffs 574 Ma[1] and sandstones Conglomerate
	Unconformity *Tilting*		
		Padarn Tuff Formation	614 ± 2 Ma[2] – 604.7 ± 1.6 Ma[1] (inherited zircons 617–638 Ma)[1]

[1]This paper.
[2]Tucker & Pharaoh (1991).

folding or unconformities between these lowest sediments and those containing Cambrian fossils it seems that in this area there was continuous sedimentation from the time of a distant volcanic episode at *c.* 570 Ma and the start of Cambrian time (Table 4).

One of the zircons in the ignimbrite of the Padarn Tuff gives a significantly older age (*c.* 638 Ma) which, from the nature of ignimbritic eruptions, is most likely to be an inherited xenocryst from the walls of the volcano. The Fachwen samples also contain older zircons and as these are fine air-fall tuffs it is again unlikely that they are derived by mechanical erosion and transport into the tuff, but give some indication of the age spectrum of the volcanic edifice. In this case several inherited ages seem the most likely explanation of the data, these ranging from 603 to 657 Ma. As these tuffs are so fine-grained the eruption could be from a volcano as far afield as the Welsh Borders or the Charnwood area and is probably not sampling the Padarn Tuff basement immediately below.

This date, of 572.5 Ma, for the Fachwen Formation also throws light on the other correlation problem in this area, i.e. the status of the four outliers of sediments and volcanics which lie unconformably on various members of the Mona Complex. The Trefdraeth Conglomerate, Carreg Onen Beds, Baron Hill Beds and the Bwlch Gwyn Tuff are undated but have sometimes been correlated with the Arfon Group. Since the blueschists on which at least one of these units rest was metamorphosed at *c.* 550 Ma (Dallmeyer & Gibbons 1987), they can not now be regarded as lateral correlatives of the Arfon Group. If the Fachwen Formation and the Llanberis Slates are essentially in conformity, it seems that there was uninterrupted sedimentation from 574 Ma until *c.* 530 Ma or later, in complete contrast to the events in Anglesey where volcanicity, sedimentation, tectonism and metamorphism in an active subduction system was taking place during this interval. Any correlation of the Fachwen with any of the sequence on Anglesey now seems unlikely.

It also now seems unlikely that the Sarn Complex should be compared with the Padarn area as the Parwyd Gneiss has yielded a Rb–Sr date of 542 ± 17 Ma (Beckinsale *et al.*, 1984). This is not now regarded as the age of the amphibolite facies metamorphism (Horák *et al.* 1996) but must indicate a considerable retrogression event in the gneissose basement in Late Precambrian–Cambrian time.

Position of England and Wales in the Avalonian–Cadomian collage

The concept of a Late Precambrian orogenic belt (Wright 1969) thus still seems to be viable, albeit in more modern (i.e. terranic) form, with oblique subduction giving rise to sedimentation, volcanicity and deformation in a wrench regime (Thorpe *et al.* 1984; Nance *et al.* 1991). A Late Precambrian volcanic arc contemporaneous over most of Central England and mainland Wales can now be regarded as proven by these more accurate dating methods. A sequence of events for the English Midlands and the Welsh Borderland (Fig. 18) now shows fairly consistent dates which can probably also be applied to South and North Wales.

The inherited zircons obtained in this study from the Charnian and Longmyndian tuffs indicate that the basement in England probably includes rocks of *c.* 2000–1000 Ma although outcrops of rocks of such an age are not known to exist anywhere in this region. Three episodes of volcanic and plutonic activity are indicated from the more recent and accurate age data, while three episodes of deformation may be deduced from the geological relationships. The earliest rocks for which there is reasonably reliable age data are the Malvernian plutons probably intruded about 700–680 Ma (Tucker & Pharaoh 1991), with geological evidence for a major deformation episode between the Malvernian and the 560 Ma Warren House Formation which overlies it.

The two later volcanic phases are now much more securely characterized by the data presented in this paper, which is supported by the more accurate of previously published dates. The older of these two phases, at *c.* 620–590 Ma, is represented by the Padarn Tuff, the Caldecote volcanics, inherited grains in the Bardon Hill Complex and probably also the Pebidian of Pembrokeshire. The Caldecote volcanics are predominantly pyroclastic rocks with some later plutons of similar geochemistry, while the Pebidian is also a varied sequence of volcanics with later plutons (Pharaoh & Gibbons 1994). The Padarn Tuff, in contrast, is a uniform acid ignimbrite.

The later period of volcanicity seems to be of quite short duration, *c.* 575–550 Ma, but only in North Wales is there clear evidence of a break between the two volcanic episodes, where the Padarn Tuff was definitely tilted before the unconformable deposition of sediments containing tuffs of the last period of volcanicity. This second volcanic episode occurs in all the areas of major volcanic outcrop, i.e. Uriconian–Longmyndian, Charnian, Warren House (Malverns) and Fachwen (Arfon). The Coomb Volcanic Group of Llangynog near Carmarthen is regarded as belonging to this group as it contains elements of the Ediacara fauna (Cope 1982; Cope & Bevins 1993). This period of volcanicity, although now seen to be contemporaneous, has been shown to belong to at least two separate geochemical provinces (Pharaoh *et al.* 1987), which are interpreted by those authors as belonging to two separate Avalonian terranes: the Charnwood Terrane regarded as a subduction-related calc-alkaline volcanic arc possibly lying on oceanic crust and the Wrekin Terrane, which has the Uriconian–Longmyndian lying on a pre-volcanic basement (Rushton Schist and Malvern Complex), with the Warren House Group (of the Malverns) a relic of a marginal basin formerly separating the Charnwood and Wrekin terranes. The contemporaneity of all these volcanics does not invalidate this interpretation. An episode of strong deformation which resulted in eastward thrusting of the Uriconian and an overturned fold some miles in amplitude at the Long Mynd seems to be of pre-Tommotian age. Since the Pebidian is also strongly cleaved (unlike the overlying Cambrian) it is likely that the South Wales area was also deformed during this late deformation episode. Transpression in a transcurrent fault regime may well explain this folding, as well as the dismembered outcrop pattern of the English and Welsh Late Precambrian. We suggest that the Charnwood Forest Anticline is also of this age, though the evidence for this is not as clear cut. The encroachment of the Cambrian sea over this area in Tommotian–Atdabanian times marks the end of the Avalonian orogenic events.

This sequence of events for the English Midlands and Wales (Fig. 18) thus represents a clear picture of multiple volcanic, plutonic and deformational episodes entirely comparable with the development of the Avalonian belt in North America (Nance *et al.* 1991) and with similar ages to events in the Cadomian of the Channel Islands and northern France (Power & Gibbons 1994). Nance *et al.* (1991) have demonstrated that in the Canadian area the most widespread volcanic sequences are the result of a magmatic arc forming around 630–600 Ma. In Britain the Caldecote Volcanics and correlatives, the Padarn ignimbrites and possibly the Pebidian are of this age. In various places in Canada an unconformable sequence of volcanics or sediments lies above these. In southern New Brunswick the Coldbrook Group (560–550 Ma, Barr *et al.* 1994) is produced by post-orogenic extension some time after the 630–600 Ma subduction-related volcanics. In SE Cape Breton Island the Main-à-Dieu Group is *c.* 563 Ma (Bevier *et al.* 1993). In the lower part of the Conception Group of Newfoundland the detritus is basically volcaniclastic and compares very closely with the Charnian in both in its volcaniclastic–sedimentological character and its metazoan fossils. The prodelta shales and alluvial red beds (St John's Group, Signal Hill of the Avalon area) which follow may well indicate a cessation in volcanicity similar to that found at the top of the Longmyndian. In SW Avalon occurs the most complete sequence of latest Precambrian to earliest Cambrian sediments which has been described in detail (Myrow & Hiscott 1993). In Avalonia this last period is ascribed to a rift or wrench regime. The overlying Cambrian from the Tommotian onwards is a platform sequence containing Acado-Baltic (Atlantic Realm) faunas.

The sequence in the Channel Islands and northern France, where the Cadomian orogeny was defined, has a similar age range with an early phase *c.* 600 Ma and a late phase *c.* 570 Ma (Power & Gibbons 1994 see also Miller *et al.* 2001). However the rock types and particularly the metamorphic and deformational phases are rather different to the Avalonian, with the 570–540 Ma phase including the St Malo migmatites—a higher grade metamorphic complex than anything seen in the English and Welsh Precambrian. The earlier phase is predominantly 600–585 Ma calc-alkaline plutons, which are probably later than the largely turbiditic Brioverian sedimentary sequence.

Nance et al. (1991) suggest that, although there is an apparent subdivision into three periods of activity, it is more likely that magmatism and deformation was semi-continuous over the area from *c.* 700–540 Ma. The data presented herein and other recent dates, however, have only strengthened the view that in the Avalonian segment there was a distinct hiatus before the 620–590 Ma phase and a strong maximum of volcanic activity (in the British Isles at least) *c.* 575–550 Ma. In North America Barr & White (1996) suggest that these later volcanics and plutons are rather sparsely developed there.

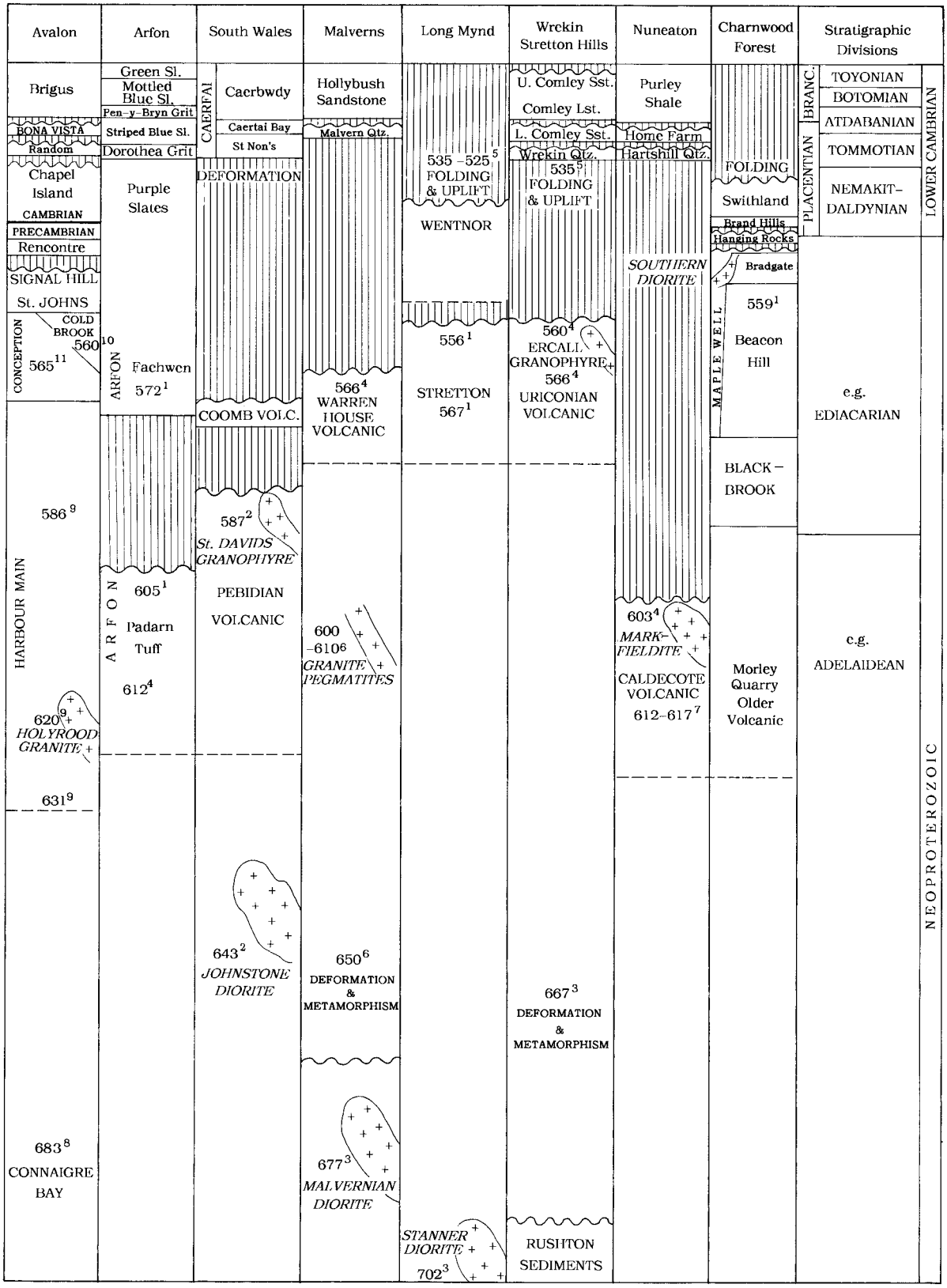


Fig. 18. Correlation of the late Precambrian and Cambrian rocks of England, Wales and Avalon. Candidate chronostratigraphic divisions after McIlroy *et al.* (1998). BRANC, Brancian Series. Lithostratigraphic names: GROUPS in caps, Formations in lower case, + *Intrusions* in italics. Precambrian–Cambrian boundary marked by thick line within Chapel Island Formation. Lower boundaries dashed where age of beginning is uncertain. [1]This paper; [2]Patchett & Jocelyn (1979); Thorpe *et al.* (1984); [3]Patchett *et al.* (1980); [4]Tucker & Pharaoh (1991); [5]see Pauley (1990) and Wright *et al.* (1993); [6]Strachan *et al.* (1996); [7]Noble *et al.* (1993); [8]Swinden & Hunt (1991); [9]Nance *et al.* (1991); [10]Barr *et al.* (1994); [11]Benus (1988).

Whether such subdivisions mark major phases in the orogenic evolution remains to be seen as more of the local sequences become more accurately dated.

Probably the most remarkable facet of this belt is the sharp change in the tectonic environment preceding the Tommotian unconformity with the stabilization of the whole area such that platform sedimentation over a cratonized Precambrian sequence is very widespread. Only in North Wales and SE Ireland does the Cambrian not appear as unconformable platform sediments. There the Cambrian is a deep water sequence, which in Arfon is apparently in conformable continuity with the beds that contain evidence of the last period of Avalonian volcanicity, which was some distance away from the site of deep water latest Precambrian–Cambrian sedimentation. The argillaceous sequence of the highest Charnian is also of Cambrian age but is of anomalous facies compared with other sequences on the craton. This Charnian area now seems to have more similarities with Arfon and SE Eire perhaps suggesting that it is very exotic to the English Midlands. However, given that there are two volcanic sequences and five or six Late Precambrian and Lower Cambrian clastic sequences in Avalon, compared with two volcanic and two or three clastic sequences in the British Avalonian, it seems that much of the evidence is missing in Britain.

Of the three superterranes of the Late Precambrian of Gibbons (1990) we have only investigated the Avalonian Superterrane. Within this superterrane the status of the Charnwood and Wrekin Terranes has been strengthened, for although they may be chemically distinct provinces, they do appear to be broadly contemporaneous in their volcanicity. But although only the Wrekin Terrane displays outcrops of an older basement, the inherited zircons found both at Charnwood (1950 and 1015 Ma) and the Longmynd (1500 Ma) indicate that both terranes are likely to be underlain by older basement. Thus the postulation that the Charnian Terrane is underlain by oceanic crust (Pharaoh & Carney 2000) is less secure. The age of the Fachwen, however, raises strong doubts of the unity of the four areas previously assigned to the Cymru Terrane (Pharaoh & Carney 2000). Arfon seems to be a terrane with a distinctive earlier volcanic episode and from 575 Ma to have been a deep water clastic basin only receiving volcanic detritus from distant areas. It shows no evidence of the late orogenic tectonic events seen in all surrounding terranes. We would suggest that the Sarn/Parwyd Gneiss sliver may be more closely allied to the Coedana Terrane of the Mona Superterrane on Anglesey and that the South Wales outcrops bear more resemblance in their sequence to the Wrekin terrane.

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